Soluzioni – Esercitazione verifica

Frazioni algebriche

Esercizio 1 – Condizioni di esistenza

a) $\dfrac{1}{3x}$

$$3x \neq 0 \implies \text{C.E.} : x \neq 0$$

b) $\dfrac{1}{4x+20}$

$$4x + 20 = 4(x+5)$$
$$4(x+5) \neq 0 \implies \text{C.E.} : x \neq -5$$

c) $\dfrac{x-1}{x^2-9}$

$$x^2 - 9 = (x-3)(x+3)$$
$$(x-3)(x+3) \neq 0 \implies \text{C.E.} : x \neq 3,\ x \neq -3$$

Esercizio 2 – Semplificazione

a) $\dfrac{x^7+x^5}{x^2+1}$

$$\frac{x^7+x^5}{x^2+1} = \frac{x^5(x^2+1)}{x^2+1}$$
$$\text{C.E.} : \forall x \in \mathbb{R} \qquad (x^2+1 > 0)$$
$$= x^5$$

b) $\dfrac{x^2-3x+2}{2x-2}$

$$\frac{x^2-3x+2}{2x-2} = \frac{(x-1)(x-2)}{2(x-1)}$$
$$\text{C.E.} : x \neq 1$$
$$= \frac{x-2}{2}$$

Esercizio 3 – Moltiplicazione e divisione

a) $\dfrac{x^3}{x^2-4} \cdot \dfrac{x+2}{x^2}$

$$\frac{x^3}{x^2-4} \cdot \frac{x+2}{x^2} = \frac{x^3}{(x-2)(x+2)} \cdot \frac{x+2}{x^2}$$
$$\text{C.E.} : x \neq -2,\ x \neq 2,\ x \neq 0$$
$$= \frac{x}{x-2}$$

b) $(x^2+2x+1) : (x^2-1)$

$$(x^2+2x+1) : (x^2-1) = (x+1)^2 : \big[(x-1)(x+1)\big]$$
$$\text{C.E.} : x \neq -1,\ x \neq 1$$
$$= \frac{(x+1)^2}{1} \cdot \frac{1}{(x-1)(x+1)}$$
$$= \frac{x+1}{x-1}$$